Grade 5 Science

Force, Motion and Energy

Booklet B: Electricity and Magnetism

Student Workbook | 2018 Virginia Science SOL 5.4

Unit 5.4: Electricity and Magnetism

Lessons 1 to 12

Review day, then the Booklet B test

Name _____

Class _____

Teacher _____

How to Use Your Workbook

Every science lesson has **three pages**, and every page shows the lesson at the top in a box that starts with **TODAY**. Find today's lesson, and you will find today's work.

1 **READ**
Read Part A and Part B with your class. Learn the Words to Know. Each bold word is important.

2 **THINK**
Read Part C. Study the picture and read its caption. Then do the Think It Through box.

3 **ANSWER**
Answer every question. The tag on the right (Look in Part B) tells you where the answer is. Go back, find the sentence, then write your answer in your own words.

Think Like a Scientist

- Read slowly. Stop at each bold word and say what it means.
- Answer in full sentences. Use the gray words to help you start.
- When you explain, first say what you know, then say why. This is called Claim, Evidence, Reasoning.
- If you are stuck, go back to the part named in the tag. The answer is always there.

Working together. Our world is full of order that we can study and understand. We take care of energy, our bodies and each other, and we use good manners when we work with a partner.

What We Learn Each Day

Each lesson takes one 40 minute class. Find today's lesson to find your pages.

Day	Lesson	What we learn	Pages
Unit 5.4: Electricity and Magnetism			
Day 1	1	Electricity and Electric Current	4 to 6
Day 2	2	Conductors and Insulators	7 to 9
Day 3	3	Parts of a Circuit	10 to 12
Day 4	4	Open and Closed Circuits	13 to 15
Day 5	5	Drawing Circuit Diagrams	16 to 18
Day 6	6	Static Electricity	19 to 21
Day 7	7	Lightning and Electrical Safety	22 to 24
Day 8	8	Electrical Energy Changes Form	25 to 27
Day 9	9	Magnets and Magnetic Fields	28 to 30
Day 10	10	Electromagnets	31 to 33
Day 11	11	Making Electromagnets Stronger	34 to 36
Day 12	12	Electricity and Magnetism at Work	37 to 39
Day 13		**Review day** (separate review sheet)	
Day 14		**Test: Booklet B**	

TODAY: Day 1 **Lesson 1: Electricity and Electric Current**

Electricity and Electric Current

GOALS

I can explain what electric charge and electric current are.
I can describe how electricity is used in daily life.

WORDS TO KNOW	
electron	a tiny particle with a negative charge that moves around the outside of an atom
electric charge	a property of tiny particles that can be positive or negative
electric current	the flow of electric charge through a material
electricity	everything that happens because of electric charges, moving or still

A Part A: Tiny Charged Particles

All matter is made of tiny particles called atoms, far too small to see. Atoms are made of even smaller parts, and some of these parts have an **electric charge**. Protons, in the center of the atom, have a positive charge. **Electrons**, which move around the outside of the atom, have a negative charge. Most of the time, an atom has the same number of protons and electrons, so the positive and negative charges balance out.

B Part B: Charges on the Move

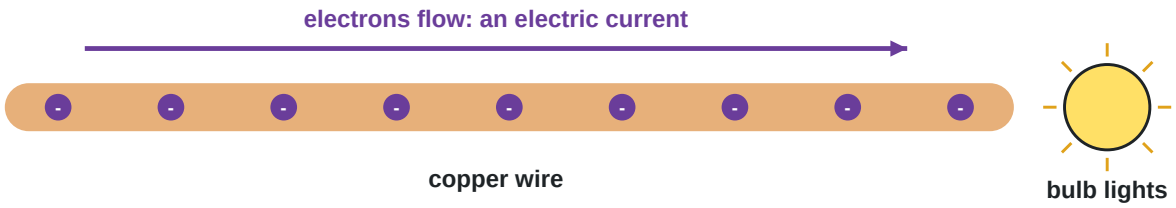
In some materials, such as copper and other metals, electrons can move easily from one atom to the next. When many electrons move in the same direction through a wire, they make an **electric current**. Electric current is the flow of electric charge. You cannot see electrons moving, but you can see what the current does: a bulb lights up, a fan spins, a speaker plays sound. **Electricity** is the word for everything that happens because of electric charges, whether they are moving or staying still.

DID YOU KNOW? The electrical signal travels rapidly through the circuit, so the light responds almost immediately. Individual electrons drift much more slowly.

TODAY: Day 1 — **Lesson 1: Electricity and Electric Current**

C Part C: Electricity in Daily Life

Electricity is one of the most useful forms of energy, because it travels quickly through wires and changes easily into other forms. Power plants make electricity, and long wires carry it to homes, schools and libraries. Inside a building, wires in the walls carry current to outlets and light switches. Batteries supply electricity to things we carry, like phones, flashlights and remote controls. Think about your morning: an alarm clock, a light, a refrigerator, a toaster. Each one needed electric current.



Electrons flowing through a wire make an electric current, and the bulb lights.

THINK IT THROUGH: Where Does the Electricity Come From?

on your own or with a partner | about 10 min

Write BATTERY, OUTLET or BUILDING SUPPLY to show where each device usually gets its electricity.

Device	Where does it come from?
A cell phone (when it is not plugged in)	
A refrigerator	
A flashlight	
The lights in our classroom	
A TV remote control	

Why do things we carry around usually run on batteries?

TODAY: Day 1 — Lesson 1: Electricity and Electric Current

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 What kind of charge does an electron have?

Look in Part A

2 Where in an atom are the protons? Where are the electrons?

Protons are

Look in Part A

3 What is electric current?

Electric current is

Look in Part B

4 In which material can electrons move easily?

Circle one: **copper** **glass** **rubber**

Look in Part B

5 We cannot see electrons. How do we know that current is flowing?

We know because

Look in Part B

6 Why is electricity such a useful form of energy?

Look in Part C

7 Name two devices you used today that needed electric current. Where did their electricity come from?

Look in Part C

TODAY: Day 2 **Lesson 2: Conductors and Insulators**

Conductors and Insulators

GOALS

I can explain the difference between conductors and insulators.
I can explain how insulators keep us safe.

WORDS TO KNOW	
conductor	a material that lets electric current flow through it easily
insulator	a material that does not let electric current flow through it easily
electrician	a person whose job is to install and fix electric wiring

A Part A: Conductors

Some materials let electric current flow through them easily. These materials are called **conductors**. Most metals are good conductors. Copper is used in most wires because it conducts very well and bends easily without breaking. Aluminum, iron, silver and gold are also conductors. Water with salt or minerals in it, like tap water and the water inside your body, can conduct electricity too.

B Part B: Insulators

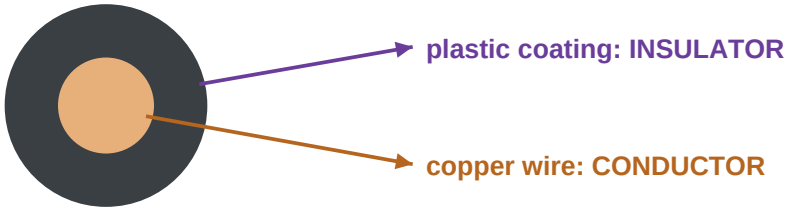
Other materials do not let electric current flow through them easily. These are **insulators**. Plastic, rubber, glass, dry wood, cloth and air are insulators. Insulators are just as important as conductors, because they keep the current where it belongs and keep people safe. Look at a power cord. Inside is a copper wire that carries the current. Outside is a plastic coating that keeps the current from reaching your hand.

DID YOU KNOW? Birds can sit safely on a single power line. The current has no reason to leave the wire and pass through the bird, because the bird is not touching anything else.

TODAY: Day 2 — **Lesson 2: Conductors and Insulators**

C Part C: Staying Safe

Because your body can conduct electricity, household current from an outlet can cause a severe electric shock and burns. Electricians use specially designed and tested insulating gloves and tools; ordinary rubber gloves or plastic handles do not make working with electricity safe. Never use an electrical device with wet hands or near water, never put anything except a plug into an outlet, and never touch a cord whose plastic coating is broken: damaged or wet insulation can lose its protection. In class, we use only teacher-approved low-voltage batteries.



A power cord: a copper conductor inside, a plastic insulator outside.

THINK IT THROUGH: Conductor or Insulator?

on your own or with a partner | about 10 min

Write CONDUCTOR or INSULATOR for each material. Use Parts A and B.

Material	Conductor or insulator?
A copper wire	
A rubber band	
A piece of aluminum foil	
A plastic spoon	
An iron nail	
A dry wooden craft stick	

TODAY: Day 2 — Lesson 2: Conductors and Insulators

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 What is a conductor?

A conductor is

_____

_____

Look in Part A

2 What is an insulator?

An insulator is

_____

_____

Look in Part B

3 Name three conductors.

_____

Look in Part A

4 Name three insulators.

_____

Look in Part B

5 In a power cord, the part that carries the current is the

Circle one: **copper wire** **plastic coating**

Look in Part B

6 Why are electrical wires covered in plastic?

_____

_____

Look in Part B

7 Why should you never use an electrical device with wet hands?

_____

_____

Look in Part C

TODAY: Day 3 **Lesson 3: Parts of a Circuit**

Parts of a Circuit

GOALS

I can name the parts of a simple circuit.
I can explain the job of each part.

WORDS TO KNOW	
circuit	a complete path that electric current can flow through
energy source	the part that supplies the energy, such as a battery
terminal	one of the two ends of a battery, marked plus (+) or minus (-)
load	the part that uses the energy, such as a bulb, a buzzer or a motor

A Part A: A Path for Current

Electric current can flow only when it has a complete path to follow. This path is called a **circuit**. The word circuit sounds like circle, and that is a good way to remember it. The path must go all the way around, from the energy source, through the wires and the devices, and back to the energy source again. If the path is not complete, there is no current.

B Part B: Three Main Parts

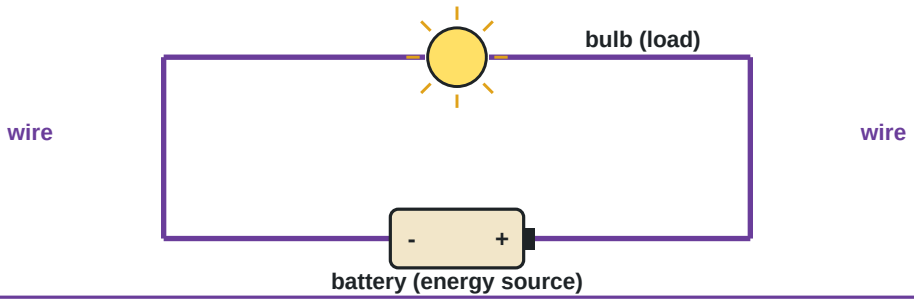
A simple circuit has three main parts. First, it needs an **energy source**, such as a battery. A battery has two ends, called **terminals**, marked plus (+) and minus (-). Second, it needs wires, usually copper covered in plastic, to carry the current. Third, it needs something that uses the energy, called a **load**, such as a light bulb, a buzzer or a small motor. In a simple circuit, one wire runs from a battery terminal to the load, and another wire runs from the load back to the other terminal.

DID YOU KNOW? A small battery for a flashlight changes stored chemical energy into electrical energy. When the chemicals are used up, the battery can no longer push current around the circuit.

TODAY: Day 3 — **Lesson 3: Parts of a Circuit**

C Part C: How a Flashlight Works

A flashlight is a circuit you can hold in your hand. The batteries are the energy source. Metal strips inside the flashlight act as the wires. The bulb is the load. When the flashlight is on, current flows from the batteries, through the metal strips and the bulb, and back to the batteries. The batteries change chemical energy into electrical energy, and the bulb changes electrical energy into light and some heat.



A simple circuit: a battery, two wires and a bulb in one complete loop.

THINK IT THROUGH: What Is Each Part's Job?

on your own or with a partner | about 10 min

Write the job of each part of a flashlight circuit.

Part	Its job in the circuit
Battery	
Metal strips (wires)	
Bulb	
Plastic case	

TODAY: Day 3 — Lesson 3: Parts of a Circuit

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 What is a circuit?
A circuit is

Look in Part A

2 How can the word circle help you remember what a circuit is?

Look in Part A

3 A simple circuit needs an energy source such as a _____, _____ to carry the current, and something that uses the energy, such as a _____.

Look in Part B

4 What are the two ends of a battery called? How are they marked?

Look in Part B

5 What is a load? Give one example.
A load is

Look in Part B

6 In a flashlight, what acts as the wires?

Look in Part C

7 Write the energy changes in a flashlight, from the batteries to the light.

Look in Part C

TODAY: Day 4 **Lesson 4: Open and Closed Circuits**

Open and Closed Circuits

GOALS

I can tell the difference between an open circuit and a closed circuit.
I can explain how a switch works.

WORDS TO KNOW	
closed circuit	a complete path, so current flows
open circuit	a path with a gap, so no current flows
switch	a device that opens or closes a circuit on purpose

A Part A: Closed Circuits

When a circuit has a complete, unbroken path, it is a **closed circuit**. Current flows all the way around, and the load works: the bulb lights or the motor spins. Every connection in a closed circuit must be metal touching metal. If even one connection is loose, the path is broken and the current stops.

B Part B: Open Circuits

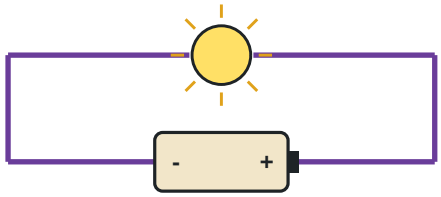
When there is a gap anywhere in the path, the circuit is an **open circuit**. In our simple battery circuits, current does not flow across an open gap, so it stops flowing everywhere in the circuit, and the bulb goes out. A gap can be made by a disconnected wire, an open switch, or a broken filament inside an incandescent bulb. Even a tiny gap, too small to see, opens a circuit. A dead battery is different: the path can stay complete, but the bulb stays off because the battery can no longer supply enough electrical energy.

DID YOU KNOW? Many homes have a breaker box. If too much current flows, a breaker switches itself off and opens the circuit to prevent a fire.

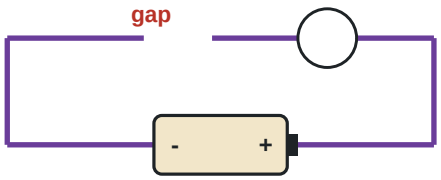
TODAY: Day 4 — **Lesson 4: Open and Closed Circuits**

C Part C: Switches

A **switch** is a device that opens or closes a circuit on purpose. Inside a light switch, a small metal piece moves. When you flip the switch on, the metal piece touches both sides and closes the circuit, so the light turns on. When you flip it off, the metal piece moves away and opens a gap, so the light goes out. Switches let us control electricity safely without ever touching the wires.



CLOSED: bulb lights



OPEN: gap, bulb is dark

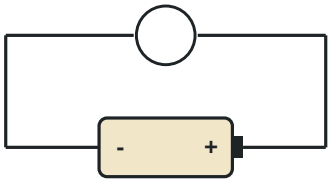
Closed: the path is complete and the bulb lights. Open: a gap stops the current.

THINK IT THROUGH: Will the Bulb Light?

on your own or with a partner | about 10 min

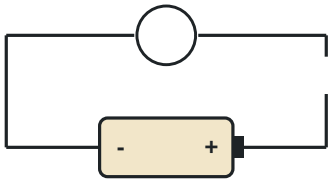
Trace each circuit with your finger. Circle LIGHTS or DOES NOT LIGHT, then write the reason.

Circuit 1



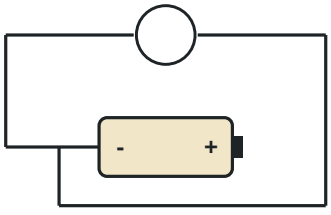
Circle: LIGHTS / DOES NOT LIGHT
Why: _____

Circuit 2



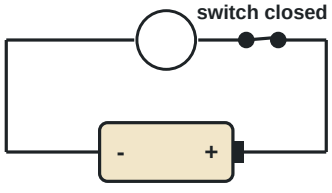
Circle: LIGHTS / DOES NOT LIGHT
Why: _____

Circuit 3



Circle: LIGHTS / DOES NOT LIGHT
Why: _____

Circuit 4



Circle: LIGHTS / DOES NOT LIGHT
Why: _____

TODAY: Day 4 — Lesson 4: Open and Closed Circuits

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 The bulb is lit. The circuit is

Circle one: **open** **closed**

Look in Part A

2 Current flows only in a _____ circuit.

Word bank: **open closed**

Look in Part A

3 Why does a bulb go out when a wire comes loose?

Look in Part B

4 Name three things that can open a circuit.

Look in Part B

5 How does a light switch on the wall work?

When I flip it on,

Look in Part C

6 Why do we use switches?

Look in Part C

7 A flashlight has new batteries, but it will not turn on. Name one thing that might be wrong.

Look in Part B

TODAY: Day 5 **Lesson 5: Drawing Circuit Diagrams**

Drawing Circuit Diagrams

GOALS

I can read the symbols in a circuit diagram.
I can draw a simple circuit diagram.

WORDS TO KNOW	
circuit diagram	a drawing that uses simple symbols to show the parts of a circuit
symbol	a simple drawing that stands for something real

A Part A: Why Engineers Use Diagrams

Electricians and engineers need to show how a circuit is built without drawing every detail. Instead of drawing a real battery, twisted wires and a glass bulb, they draw a **circuit diagram**. A circuit diagram uses simple **symbols**. Anyone who knows the symbols can read the diagram, in any country and in any language. A circuit diagram is like a map of the path the current will take.

B Part B: The Symbols

A battery is drawn as a long line and a short line side by side. The long line is the plus terminal. A wire is drawn as a straight line. A bulb is drawn as a circle with an X inside. A switch is drawn as a line with a small piece on a hinge. When the switch is open, the piece is drawn lifted up, leaving a gap. Wires are always drawn with straight lines and square corners, which keeps the diagram neat and easy to read.

DID YOU KNOW? Electricians use circuit diagrams to plan all the wiring for a new house before a single wire is put inside the walls.

TODAY: Day 5 — **Lesson 5: Drawing Circuit Diagrams**

C Part C: Reading a Diagram

To read a circuit diagram, put your finger on the battery and trace the path. If your finger can travel all the way around and come back to the battery without lifting up, the circuit is closed and the bulb will light. If your finger reaches a gap, such as an open switch or a missing wire, the circuit is open and the bulb will not light. Tracing with a finger works for any diagram, even a big one.



The four circuit symbols you need to know.

THINK IT THROUGH: Draw a Circuit Diagram

on your own or with a partner | about 10 min

In the box, draw a circuit diagram with ONE battery, ONE bulb and ONE switch that is CLOSED. Use the symbols. Use straight lines and square corners.

Will your bulb light? Explain how you know.

TODAY: Day 5 — Lesson 5: Drawing Circuit Diagrams

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 What is a circuit diagram?

A circuit diagram is

Look in Part A

2 Why do engineers use symbols instead of real pictures?

Look in Part A

3 Describe the symbol for each part.

Look in Part B

Battery	
Bulb	
Wire	
Open switch	

4 In the battery symbol, which line is the plus terminal?

Look in Part B

5 How can you tell from a diagram whether the bulb will light?

Look in Part C

6 A diagram shows a switch drawn lifted up, with a gap. The bulb

Circle one: **will light** **will not light**

Look in Part C

7 Why are wires drawn with straight lines and square corners?

Look in Part B

TODAY: Day 6 **Lesson 6: Static Electricity**

Static Electricity

GOALS

I can explain how static electricity builds up.
I can explain why charged objects attract or repel.

WORDS TO KNOW	
static electricity	a buildup of electric charge on the surface of an object
neutral	having equal amounts of positive and negative charge
attract	to pull toward
repel	to push away

A Part A: Charges Build Up

Have you ever rubbed a balloon on your hair and watched your hair rise toward the balloon? That is **static electricity**, a buildup of electric charge on the surface of an object. Most objects are **neutral**, meaning they have the same amount of positive and negative charge. When two materials rub together, some electrons move from one material to the other. The material that gains electrons becomes negatively charged. The material that loses electrons becomes positively charged. Rubbing does not make new charge. It only moves electrons that were already there.

B Part B: Attract and Repel

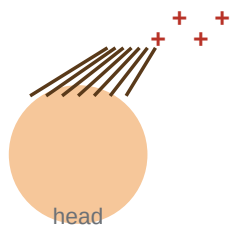
Charged objects push and pull on each other. Opposite charges **attract**, or pull together. Like charges, two positives or two negatives, **repel**, or push apart. After rubbing, the balloon has extra electrons and a negative charge, and your hair has lost electrons and has a positive charge. Opposite charges attract, so your hair rises toward the balloon. Every hair has the same positive charge, so the hairs also repel each other and spread apart.

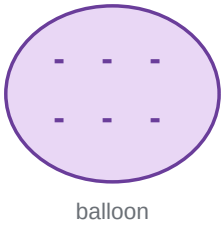
DID YOU KNOW? Static electricity can damage computer parts, so people who build computers often wear a special strap that safely carries away any built up charge.

TODAY: Day 6 **Lesson 6: Static Electricity**

C Part C: Static All Around Us

You can see static electricity in many places. Socks cling together when they come out of the dryer. You may feel a small shock when you touch a metal doorknob after walking across a carpet. A charged balloon can even stick to a wall. Static electricity is different from the current in a circuit. In a circuit, charge flows steadily along a path. With static electricity, charge builds up in one place and stays there until it jumps away in a quick spark.





balloon: NEGATIVE (gained electrons)

hair: POSITIVE (lost electrons)

opposite charges attract

After rubbing, the balloon is negative and the hair is positive. Opposites attract.

THINK IT THROUGH: Attract or Repel?

on your own or with a partner | about 10 min

Write ATTRACT or REPEL for each pair of charges.

Pair of charges	Attract or repel?
positive and negative	
negative and negative	
positive and positive	
a negative balloon and positive hair	

True or false: rubbing a balloon creates new charge. Explain.

TODAY: Day 6 — Lesson 6: Static Electricity

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 What is static electricity?

Static electricity is

Look in Part A

2 How does rubbing two materials make them charged?

Look in Part A

3 Opposite charges _____. Like charges _____.

Word bank: **attract repel**

Look in Part B

4 Why does your hair rise toward a rubbed balloon?

Look in Part B

5 What does neutral mean?

Look in Part A

6 Give two examples of static electricity in daily life.

Look in Part C

7 How is static electricity different from current in a circuit?

Look in Part C

TODAY: Day 7 **Lesson 7: Lightning and Electrical Safety**

Lightning and Electrical Safety

GOALS

I can explain how lightning forms.
I can describe how people stay safe from lightning.

WORDS TO KNOW	
discharge	a sudden flow of built up charge from one place to another
lightning	a giant spark of static electricity in the sky
lightning rod	a metal rod that gives lightning a safe path to the ground

A Part A: Sparks

Built up static charge does not stay forever. When enough charge builds up, it can jump to another object in a quick flow called a **discharge**. The small shock you feel from a doorknob is a tiny discharge: electrons jump between your hand and the metal. In a dark room you might even see a small spark or hear a crackle when you pull off a wool sweater.

B Part B: Lightning

Lightning is the same thing on a giant scale. Inside a storm cloud, tiny pieces of ice and drops of water crash into each other as strong winds carry them up and down. These collisions move electric charges, and huge amounts of charge build up in different parts of the cloud. When the charge gets large enough, it discharges in an enormous spark between parts of the cloud, between clouds, or between the cloud and the ground. The spark heats the air so quickly that the air expands with a boom, which we hear as thunder.

DID YOU KNOW? A single bolt of lightning is hotter than the surface of the sun, and it heats the air around it in a tiny fraction of a second.

TODAY: Day 7 — **Lesson 7: Lightning and Electrical Safety**

C Part C: Staying Safe in a Storm

Lightning is extremely dangerous, so people protect themselves and their buildings. Many tall buildings have a **lightning rod** on top. It is a metal rod joined by a thick wire to the ground. If lightning strikes, the current flows safely through the rod and wire into the ground instead of through the building. During a thunderstorm, go inside a substantial enclosed building or a fully enclosed, metal-topped car with the windows closed (not a shed, picnic shelter or convertible). Stay away from water and tall trees, and wait 30 minutes after the last thunder.



Charge builds up in a storm cloud. A lightning rod gives the discharge a safe path.

THINK IT THROUGH: Safe or Not Safe?

on your own or with a partner | about 10 min

A thunderstorm has started. Write SAFE or NOT SAFE for each choice.

What a person does	Safe or not safe?
Keeps swimming in an outdoor pool	
Stands under a tall tree to stay dry	
Goes inside the school building	
Waits in a closed, metal-topped car	
Keeps playing soccer on an open field	

Pick one NOT SAFE choice and explain why it is dangerous.

TODAY: Day 7 — Lesson 7: Lightning and Electrical Safety

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 What is a discharge?

A discharge is

_____

_____

Look in Part A

2 Why do you sometimes feel a shock when you touch a doorknob?

_____

_____

Look in Part A

3 What causes charge to build up inside a storm cloud?

_____

_____

Look in Part B

4 What is thunder?

Thunder is

_____

Look in Part B

5 Lightning is a giant spark of

Circle one: **static electricity** **magnetism** **sound**

Look in Part B

6 How does a lightning rod protect a building?

_____

_____

Look in Part C

7 Name two things you should do to stay safe in a thunderstorm.

_____

_____

Look in Part C

TODAY: Day 8 **Lesson 8: Electrical Energy Changes Form**

Electrical Energy Changes Form

GOALS

I can name the forms electrical energy changes into.
I can describe ways to use electricity wisely.

WORDS TO KNOW	
motor	a device that changes electrical energy into mechanical energy (motion)
heating element	a part that gets hot when current flows through it
energy efficient	using less energy to do the same job

A Part A: One Form, Many Uses

Electrical energy is so useful because it can be changed into almost any other form. A lamp changes it into radiant energy. A toaster, an oven and a hair dryer change it into thermal energy using a **heating element**, a part that gets hot when current flows through it. A speaker and a doorbell change it into sound energy. A fan and a blender use a **motor**, which changes electrical energy into mechanical energy.

B Part B: More Than One Output

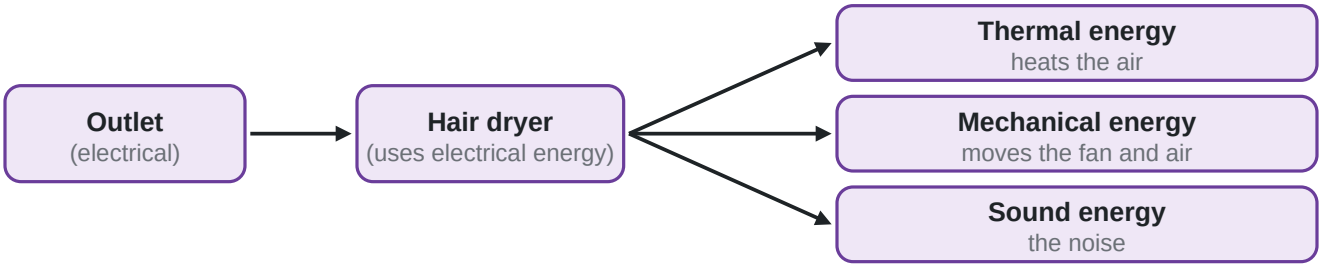
Most devices change electrical energy into more than one form. A hair dryer makes thermal energy to dry your hair, mechanical energy to spin its fan, and sound energy you can hear. A television makes radiant energy on its screen and sound energy from its speakers. A computer makes light, sound and some heat. Remember the law of conservation of energy: the energy coming out, in all its forms, adds up to the electrical energy that went in.

DID YOU KNOW? An LED bulb can last more than 15 years of normal use, much longer than an old style bulb.

TODAY: Day 8 — **Lesson 8: Electrical Energy Changes Form**

C Part C: Using Electricity Wisely

Most electricity in Virginia is made at large power plants, and many of them burn fuels such as natural gas. Using electricity wisely saves energy and money. We can turn off lights and devices when we leave a room, unplug chargers we are not using, and choose **energy efficient** devices, such as LED light bulbs, which give the same light while using much less electricity.



A hair dryer changes electrical energy into thermal, mechanical and sound energy.

THINK IT THROUGH: Device Detective

on your own or with a partner | about 10 min

Write the forms of energy each device changes electrical energy into. Many have more than one.

Device	Electrical energy becomes
Ceiling lights	
Projector	
Electric pencil sharpener	
Computer	
Electric stove	

TODAY: Day 8 — Lesson 8: Electrical Energy Changes Form

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 Toaster: electrical energy becomes _____. Lamp: electrical energy becomes _____. Doorbell: electrical energy becomes _____.

Look in Part A

2 What does a motor do?

A motor

_____

Look in Part A

3 Name one device that uses a heating element.

_____

Look in Part A

4 A hair dryer makes three forms of energy. Name them.

_____

Look in Part B

5 A speaker changes electrical energy into

Circle one: **sound energy** **chemical energy** **magnetic energy**

Look in Part A

6 Why do we use electrical energy so much?

_____

_____

Look in Part A

7 Give two ways to use electricity wisely.

_____

_____

Look in Part C

TODAY: Day 9 **Lesson 9: Magnets and Magnetic Fields**

Magnets and Magnetic Fields

GOALS
I can describe what magnets attract.
I can explain magnetic poles and magnetic fields.

WORDS TO KNOW	
magnet	an object that attracts iron and some other metals
magnetic pole	an end of a magnet, north or south, where its force is strongest
magnetic field	the area around a magnet where its force acts

A Part A: Magnets

A **magnet** is an object that attracts iron and a few other metals, such as nickel and cobalt. Steel, which is mostly iron, is attracted too, which is why magnets stick to many refrigerator doors. Magnets do not attract every metal. Aluminum foil, copper wire and most coins are not attracted. Magnets do not attract wood, plastic, glass or paper either. A magnet can pull on an iron object without touching it.

B Part B: Poles

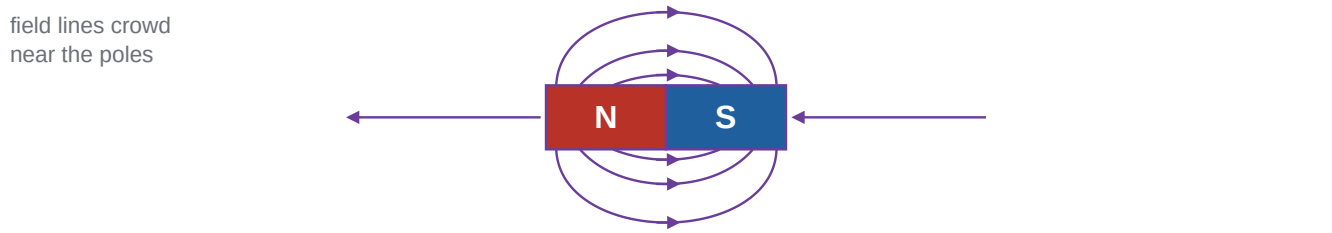
Every magnet has two ends called **magnetic poles**: a north pole and a south pole. The magnetic force is strongest at the poles. When two magnets come close, opposite poles attract, so north pulls on south. Like poles repel, so north pushes away north, and south pushes away south. This rule sounds a lot like the rule for electric charges, and that is a clue that electricity and magnetism are related.

DID YOU KNOW? Some animals, such as sea turtles and some birds, can sense Earth's magnetic field and use it to find their way on long journeys.

TODAY: Day 9 — **Lesson 9: Magnets and Magnetic Fields**

C Part C: Magnetic Fields

The area around a magnet where its force acts is called a **magnetic field**. You cannot see a magnetic field, but you can see what it does. If you sprinkle tiny bits of iron around a magnet, they line up in curved lines that loop from one pole to the other. The lines crowd closest together near the poles, where the field is strongest. Earth itself acts like a giant magnet, which is why a compass needle, a small magnet, points north.



Field lines leave the north pole and enter the south pole. They crowd together near the poles, where the field is strongest.

THINK IT THROUGH: Attract, Repel or Nothing?

on your own or with a partner | about 10 min

Write ATTRACT, REPEL or NOTHING for each pair.

Pair	What happens?
North pole near a south pole	
North pole near a north pole	
A magnet near an iron nail	
A magnet near aluminum foil	
South pole near a south pole	
A magnet near a plastic spoon	

TODAY: Day 9 — Lesson 9: Magnets and Magnetic Fields

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 Name three metals that a magnet attracts.

Look in Part A

2 Which one is attracted to a magnet?

Circle one: **aluminum foil** **a steel paper clip** **copper wire**

Look in Part A

3 What are the two poles of a magnet called?

Look in Part B

4 Opposite poles _____. Like poles _____.

Word bank: **attract repel**

Look in Part B

5 What is a magnetic field?

A magnetic field is

Look in Part C

6 Where is a magnet's field strongest? How can you tell?

Look in Part C

7 Why does a compass needle point north?

Look in Part C

TODAY: Day 10 **Lesson 10: Electromagnets**

Electromagnets

GOALS
I can explain how electric current makes a magnetic field.
I can describe an electromagnet and why it is useful.

WORDS TO KNOW	
electromagnet	a coil of wire that produces a magnetic field when electric current flows through it; an iron core makes the field stronger
coil	wire wound around and around in loops
core	the object inside a coil, often an iron nail or bar

A Part A: Current Makes a Magnetic Field

About 200 years ago, scientists discovered something amazing. When electric current flows through a wire, it makes a magnetic field around the wire. A compass needle placed near the wire swings to a new direction the moment the current is switched on, and swings back when it is switched off. This discovery showed that electricity and magnetism are closely connected.

B Part B: Building an Electromagnet

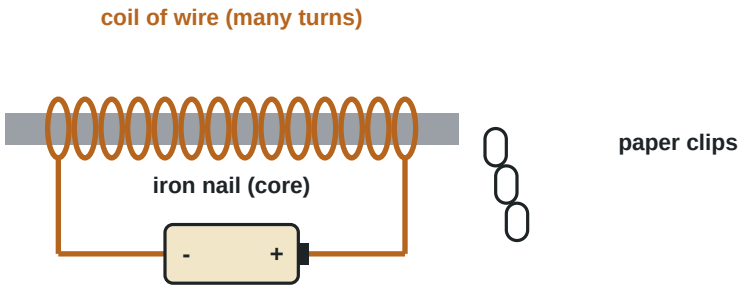
One straight wire makes only a weak magnetic field. But if the wire is wound into a **coil**, with many loops side by side, the fields of all the loops add together. A coil of wire with current flowing through it is an **electromagnet**, even with nothing inside it. If the coil is wrapped around an iron **core**, such as an iron nail, the iron becomes magnetized and the electromagnet gets much stronger. When the ends of the wire are joined to a battery, the nail can pick up steel paper clips.

DID YOU KNOW? A doorbell has an electromagnet inside. When you press the button, the electromagnet pulls a small metal striker that hits the bell.

TODAY: Day 10 **Lesson 10: Electromagnets**

C Part C: A Magnet You Can Switch Off

An electromagnet has one big advantage over an permanent magnet: it can be turned on and off. While the current flows, it is a magnet. When the current stops, most of its magnetism disappears (the iron core may keep a little), and the steel it was holding drops. That makes electromagnets perfect for junkyard cranes that lift old cars and then drop them in a new place. Electromagnets are also inside doorbells, speakers and electric motors.



An electromagnet: a coil of wire around an iron nail, joined to a battery.

THINK IT THROUGH: Permanent Magnet or Electromagnet? on your own or with a partner | about 10 min

Write YES or NO in each box.

Question	Permanent magnet	Electromagnet
Can it be switched on and off?		
Does it need electric current?		
Can its strength be adjusted by changing the electric current?		

Why does a junkyard crane use an electromagnet instead of a permanent magnet?

TODAY: Day 10 — Lesson 10: Electromagnets

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 What happens around a wire when current flows through it?

Look in Part A

2 What did this discovery show?

Look in Part A

3 What is an electromagnet?

An electromagnet is

Look in Part B

4 Why is a coil stronger than one straight wire?

Look in Part B

5 What does the iron core do?

Look in Part B

6 True or false: when the current stops, most of an electromagnet's magnetism disappears.

Circle one: **TRUE** **FALSE**

Look in Part C

7 Name two devices that use electromagnets.

Look in Part C

TODAY: Day 11 **Lesson 11: Making Electromagnets Stronger**

Making Electromagnets Stronger

GOALS

I can name the things that change how strong an electromagnet is.
I can explain when a thicker wire can make a stronger electromagnet.

WORDS TO KNOW	
turns	the number of loops of wire in a coil
wire gauge	a number that tells how thick a wire is; a smaller number means a thicker wire

A Part A: More Turns

Engineers can control how strong an electromagnet is. One way is to change the number of **turns**, the loops of wire in the coil. To compare fairly, keep the current, the core and the coil length the same. Coil length means the end-to-end length of the wound coil, not the total length of wire. Under these controlled conditions, more turns produce a stronger magnetic field, because the fields of more loops add together, and will generally allow the electromagnet to pick up more paper clips.

B Part B: More Current and an Iron Core

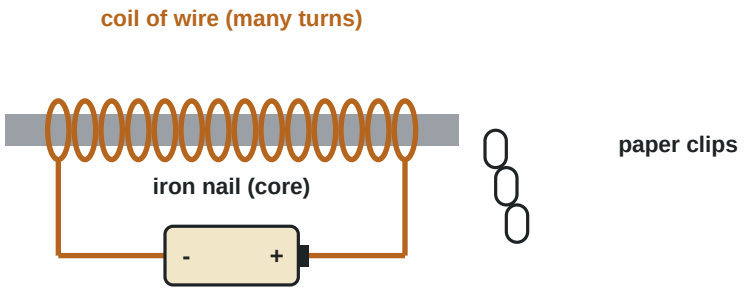
A second way is to increase the current, for example by using a battery that can push more current. More current through the coil makes a stronger magnetic field. Students must never use wall outlets or add batteries without a teacher: household electricity can cause severe electric shock and burns, so classroom investigations use only teacher-approved low-voltage batteries. A third way is the core. An iron core makes the electromagnet far stronger than a coil with no core, or a coil around wood or plastic.

DID YOU KNOW? The electromagnets inside an MRI machine at a hospital are thousands of times stronger than a refrigerator magnet.

TODAY: Day 11 — **Lesson 11: Making Electromagnets Stronger**

C Part C: Thicker Wire

The thickness of the wire matters too. Wire thickness is described by a number called the **wire gauge**. In the American Wire Gauge (AWG) system, a smaller number means a thicker wire, so 22 AWG wire is thicker than 30 AWG wire. For two wires of the same metal and length, the thicker wire has less resistance, so with the same battery more current can flow, and more current makes a stronger electromagnet. Engineers choose the turns, the current, the core and the wire gauge to build exactly the electromagnet they need.



Turns, current, the core and the wire gauge can all change the strength.

THINK IT THROUGH: Read the Test Data

on your own or with a partner | about 10 min

SAMPLE DATA for practice: made-up class results, not your own test.

What the class used	Paper clips picked up
Turns comparison: same current, iron core, 22 AWG wire and end-to-end coil length	
10 turns, 22 AWG wire	4
20 turns, 22 AWG wire	9
30 turns, 22 AWG wire	13
Wire-thickness comparison: same battery, core, number of turns, wire metal, wire length and winding; the current could change	
20 turns, 22 AWG wire (thicker)	9
20 turns, 30 AWG wire (thinner)	5

Turns comparison: what was the independent variable, and what pattern do you see? **Wire-thickness comparison:** which wire made the stronger electromagnet, and why?

TODAY: Day 11 — Lesson 11: Making Electromagnets Stronger

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 Name four things that change how strong an electromagnet is.

Look in Part A, B and C

2 Why do more turns usually make a stronger electromagnet?

Look in Part A

3 Which core makes the strongest electromagnet?

Circle one: **iron** **wood** **plastic**

Look in Part B

4 Why must students never use wall outlets to make electromagnets?

Look in Part B

5 In the American Wire Gauge (AWG) system, a _____ number means a thicker wire.

Word bank: **smaller bigger**

Look in Part C

6 Which wire is thicker?

Circle one: **22 AWG** **30 AWG**

Look in Part C

7 Why can a thicker wire of the same metal and length make a stronger electromagnet?

Look in Part C

TODAY: Day 12 **Lesson 12: Electricity and Magnetism at Work**

Electricity and Magnetism at Work

GOALS
I can explain what motors and generators do.
I can design a device that uses an electromagnet.

WORDS TO KNOW	
motor	a device that uses electricity and magnets to make motion
generator	a device that uses motion and magnets to make electricity
power plant	a place where large generators make electricity

A Part A: Motors

Inside every electric fan, blender and washing machine is a **motor**. A motor uses an electromagnet and permanent magnets that push and pull on each other. As the current flows, the electromagnet is attracted and repelled over and over, which makes a shaft spin. A motor changes electrical energy into mechanical energy.

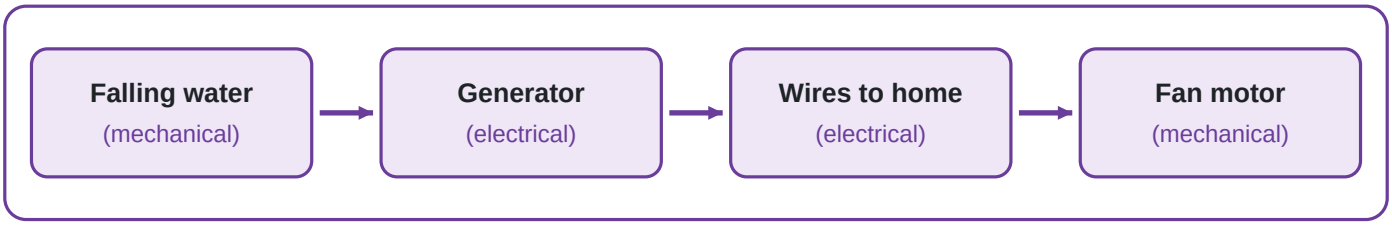
B Part B: Generators

A **generator** does the opposite job. When a magnet and a coil of wire move past each other, an electric current flows in the wire. A **power plant** uses huge generators. Steam, falling water or wind spins the generators, and the moving magnets and coils produce the electricity that travels to our homes. A generator changes mechanical energy into electrical energy. The scientist Michael Faraday discovered how to do this almost 200 years ago.

C Part C: Engineers at Work

Engineers use electromagnets to solve real problems. Hospitals use giant electromagnets in MRI machines to take pictures of the inside of the body. Some high speed trains float above their tracks on powerful electromagnets, so there is almost no friction. Recycling centers use electromagnetic cranes to pull steel cans away from aluminum cans, because steel is attracted to magnets and aluminum is not.

TODAY: Day 12 — **Lesson 12: Electricity and Magnetism at Work**



A generator makes electricity from motion. A motor makes motion from electricity.

DESIGN IT: My Electromagnet Device

Choose ONE problem. Design a device on paper that uses an electromagnet to solve it. Tick one box.

[ ] A crane that lifts steel cans out of the recycling and drops them in a bin

[ ] A door lock that opens when a button is pressed

[ ] A toy that picks up and drops paper clips

1. The problem in my own words:

2. Two criteria (what my device must do):

3. One constraint (a limit, such as cost, size or materials):

4. My sketch. Label every part.

5. My energy chain. Write the form of energy in each box.

6. How I would test my prototype:

TODAY: Day 12 — Lesson 12: Electricity and Magnetism at Work

Name: _____

Answer each question. The tag on the right tells you which part to look in. Gray words help you start.

1 What does a motor do?

A motor

Look in Part A

2 What does a generator do?

A generator

Look in Part B

3 Name two things that can spin the generators at a power plant.

Look in Part B

4 Why can a recycling center use an electromagnet to separate steel cans from aluminum cans?

Look in Part C

5 Name one way hospitals or trains use electromagnets.

Look in Part C